PROGRAMING FUNDAMENTAL PROJECT:

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ECAT Online Examination Portal Report.

Problem Statement

Conducting examinations manually can be time-consuming and difficult to manage. Teachers have to prepare questions, check answers, calculate marks, and maintain student records separately. This process may lead to errors and requires extra effort. Therefore, there is a need for a computerized examination system that can manage exams, calculate results automatically, and keep student records in an organized manner. The ECAT Online Examination Portal is developed to make the examination process easier, faster, and more efficient.

Objective

• To develop an automated ECAT Online Examination Portal for conducting and managing examinations efficiently.
• To generate accurate results and performance reports automatically while reducing manual effort and errors.

Methodology

The ECAT Online Examination Portal was developed using Python programming language. The system consists of two major modules: the Admin Portal and the Student Portal. Both modules are protected through a login system to ensure authorized access. The Admin Portal is responsible for managing examination activities, while the Student Portal is used for attempting examinations.
A collection of multiple-choice questions is stored using Python dictionaries inside a list data structure. Each question contains a subject name, question statement, four options, and the correct answer. The administrator can view existing questions, add new questions, delete unwanted questions, and monitor the total number of available questions in the system.
Students can access the examination through the Student Portal after successful login. Before starting the test, they can view the examination rules and marking scheme. During the examination, students are provided with options to answer questions, skip questions, move to the next question, return to previous questions, or submit the exam before completion. All responses entered by the student are stored temporarily during the examination process.
Once the examination is submitted, the system automatically compares student responses with the correct answers stored in the question bank. Marks are awarded according to the predefined marking criteria where correct answers receive positive marks, incorrect answers receive negative marks, and skipped questions receive no marks. The total score, percentage, and grade are then calculated automatically.

The generated results are temporarily stored in a list during program execution and can be viewed by the administrator. The administrator can view all student results, search for a specific student's detailed result using the roll number, and review the answers attempted by the student. In addition, the system calculates overall class statistics such as the highest score, lowest score, and average score, providing a clear overview of student performance.

Results

The developed system successfully performs user authentication, question management, online examination handling, automatic score calculation, percentage generation, grading, and result storage during program execution. It enables administrators to manage examinations effectively and allows students to attempt exams in an organized environment. The system also provides detailed result analysis and class performance statistics.

Conclusion

The ECAT Online Examination Portal successfully automates the examination process and minimizes manual work involved in conducting and evaluating tests. The system provides accurate result calculation, efficient question management, and reliable performance evaluation. It serves as a practical solution for educational institutions and demonstrates the effective use of Python in developing academic management applications.